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Plan of Internship program

Research Summer Internship IIT Palakkad - 45days [6 weeks 3/6/2024 - 16/7/2024]			
Topic	Power Electronic Circuits - Numerical Simulations	Dept	EE - Electrical Engineering
		Faculty	Dr. Naga Brahendra - Assistant Professor
Weeks	Details	Weeks	Details
Week - 1	Introduction to power electronics. Software setup[LT spice, Altium Pcb design]. Voltage pulse generator and MOSFET charactries. Power electronics converters _circuit practice[12 Examples] [Learnings - Converters and EDA Tools]	Week - 4	NCP 51820 Gate Driver _ ON semi GaN 3R2_100 CBE _ MOSFET _ Nexperia Data sheet, Symbol and Component creation in LT Spice and Altium PCB design [learnings - Creating of Libraries in EDA Tools and Gate driver and MOSFET]
Week - 2	Power Electronics Circuits design and Numerical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Buck 2.Boost and 3.Buck Boost [Learnings - Calculation and circuit design and simulations in LTspice]	Week - 5	Design and Simulating of Synchronous Buck boost converter Schmetic design and PCB layout component assembly and routing [Learnings - Altium PCB desining]
Week - 3	Power Electronics Circuits design and Numerical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Forward [CCM & DCM] 2. Flyback 3.Pushpull [Learnings - Calculation and circuit design and simulations in LTspice]	Week - 6	

Out comes

- * Power Electronics Converters
- * Calcuation and formula based sheets creation
- * Circuit Simulation in LT Spice Tool
- * Gate Drivers _ NCP51820 _ Onsemi
- * GaN MOSFET GAN3R2-100CBE _ Nexperia
- * EDA Tools - Altium PCB design
- * Skills - LT spice simulations, Numerical calculations and Altium Schematic and PCB designing.